Type MS · Explain how insulin and glucagon control blood glucose

Answer 1 of 11 · Question 8 part a · 3 marks

2024 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

8	(a)	any three from • released from pancreas; • when glucose levels rise; • causes/stimulates conversion of glucose to glycogen; • in liver/muscles;		(3)
	(b) (i)	• vasodilation; • more blood flows at/near the surface of the skin; • heat (from blood) radiated (away from body)/increased heat loss; • hairs lie flat; • less air trapped; • (more) heat radiated;		(3)
	(ii)	vasoconstriction/blood vessels in the skin narrowed shivering/increased metabolic rate/hairs stand on end;		(1)
	(iii)	• hypothalamus detects rise/fall/change in body temperature; • (named) mechanism reverses the change/brings back to set point/normal;	e.g. vasodilation, increase in sweating	(2)
	(c)	• enzymes lose kinetic energy; • enzyme activity decreases/works slowly; • less enzyme substrate complexes formed/ less collisions between enzyme and substrate/binding with substrate; • rate of chemical reactions/metabolism slow down; • reference to symptom of hypothermia e.g. death/weak pulse/slower heart/breathing rate;		(5)

Total for Question 8 = 14 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 2 of 11 · Question 7 part b.i · 3 marks

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

Question number					
7	(a)	(i)	• 67.5 ÷ 48.3 (1) • 1:1.4 (1)	Full marks for correct final answer	2
		(ii)	Any two from • more males than females are overweight in all countries (1) • lowest percentage of people between the ages of 18-24 in all countries are overweight (1) • increased in all countries 18 - 64 (1)		2
		(iii)	sample size too small (1) not all overweight people identified/sample not representative (1)		2
	(b)	(i)	Any three from: • pancreas (1) • releases insulin (1) • reference to the liver taking up glucose (1) • glucose converted to glycogen (1)	R sugar R breaks down glucose	3
		(ii)	Any two from: • high blood glucose increases the concentration of the blood (1) • water moves out of cells (1) • by osmosis (1) • excess water in blood excreted in urine (1)		2

Total for Question 7 = 11 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 3 of 11 · Question 4 part b.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
4 (a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
	(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
	(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100;}{22092}$ = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
	(ii)	0.9 x 3.8 = 3.4(2) million;		1
(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 3 of 11 · Question 4 part b.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 4 of 11 · Question 4 part b.i · 1 mark

2022 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
4	(a)	• maintain a constant/stable/optimum; • internal environment; • despite changes in the external environment;		3
	(b) (i)	B; (glucagon) *A is secreted by the kidneys* *C is insulin and reduces blood glucose* *D is testosterone and it is secreted by the testes .*		1
	(ii)	C; (insulin) *A is secreted by the kidneys* *C is glucagon and increases blood glucose* *D is testosterone and it is secreted by the testes.*		1
	(iii)	P = glycogen; Q = glucose; R = fat/lipid; S = respiration;		4
	(c)	• low levels of blood glucose; • reduce insulin release; • until balance/correct/normal level is reached/increases blood glucose; • secretion of insulin starts again/glucagon stopped;	accept reverse argument for glucagon	4

Total : 14 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 4 of 11 · Question 4 part b.i · 1 mark

2022 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 5 of 11 · Question 4 part b.ii · 1 mark

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a)	• maintain a constant/stable/optimum; • internal environment; • despite changes in the external environment;		3
(b) (i)	B; (glucagon) *A is secreted by the kidneys* *C is insulin and reduces blood glucose* *D is testosterone and it is secreted by the testes .*		1
(ii)	C; (insulin) *A is secreted by the kidneys* *C is glucagon and increases blood glucose* *D is testosterone and it is secreted by the testes.*		1
(iii)	P = glycogen; Q = glucose; R = fat/lipid; S = respiration;		4
(c)	• low levels of blood glucose; • reduce insulin release; • until balance/correct/normal level is reached/increases blood glucose; • secretion of insulin starts again/glucagon stopped;	accept reverse argument for glucagon	4

Total : 14 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 5 of 11 · Question 4 part b.ii · 1 mark

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 6 of 11 · Question 4 part b.iii · 4 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

number				
4	(a)	• maintain a constant/stable/optimum; • internal environment; • despite changes in the external environment;		3
	(b) (i)	B; (glucagon) *A is secreted by the kidneys* *C is insulin and reduces blood glucose* *D is testosterone and it is secreted by the testes .*		1
	(ii)	C; (insulin) *A is secreted by the kidneys* *C is glucagon and increases blood glucose* *D is testosterone and it is secreted by the testes.*		1
	(iii)	P = glycogen; Q = glucose; R = fat/lipid; S = respiration;		4
	(c)	• low levels of blood glucose; • reduce insulin release; • until balance/correct/normal level is reached/increases blood glucose; • secretion of insulin starts again/glucagon stopped;	accept reverse argument for glucagon	4

Total : 14 marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 6 of 11 · Question 4 part b.iii · 4 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 7 of 11 · Question 7 part b · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Explain how insulin and glucagon control blood glucose

Answer 7 of 11 · Question 7 part b · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Explain how insulin and glucagon control blood glucose

Answer 7 of 11 · Question 7 part b · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 8 of 11 · Question 3 part c.i · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Explain how insulin and glucagon control blood glucose

Answer 8 of 11 · Question 3 part c.i · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 9 of 11 · Question 4 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Explain how insulin and glucagon control blood glucose

Answer 9 of 11 · Question 4 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 10 of 11 · Question 4 part a.iii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Explain how insulin and glucagon control blood glucose

Answer 10 of 11 · Question 4 part a.iii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how insulin and glucagon control blood glucose

Answer 11 of 11 · Question 3 part c.ii · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Explain how insulin and glucagon control blood glucose

Answer 11 of 11 · Question 3 part c.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks